

An Explicit Windowed Sparsity Bound for Divergent $3x + 1$ Orbits, with Logarithmic-Floor Exclusion beyond the Harmonic Barrier

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24 August 2026

Abstract

Garcia and Tal proved in 1999 that every infinite orbit of the $3x + 1$ map has Banach density zero. We give a short self-contained proof of an explicit quantitative form: for every infinite aperiodic orbit $\mathcal{O}$,

$$\#(\mathcal{O} \cap [a, a + X)) \leq C X^\beta \log(2X) \quad \text{uniformly in } a,$$

for every $\beta > \beta^* = r_H \log_2 3 = 0.96538\dots$, where $r_H = 0.609089\dots$ is Rozier's critical ones-ratio. Three consequences follow for the odd-only (Syracuse) map with valuation drift $g_n = n \log_2 3 - A_n$. First, every divergent orbit has summable reciprocals, $\sum_n 1/y_n < \infty$. Second, no divergent orbit satisfies $g_n \leq B \log_2 n + O(1)$ eventually for any $B \leq 1$. Third, the occupation bound $\#\{n : g_n \leq G\} = O(2^{\beta G})$ extends the exclusion to every $B < 1/\beta^* = 1.03585\dots$. This passes the threshold $8/9$ recently proved by De Jesús via congruence-packing and shown there to be terminal for that method, and it settles, in a stronger form than requested, the graded harmonic-thinness requirement posed in his Open Problem 12.8: the harmonic mass of a divergent orbit is not merely $(\frac{1}{3} - \eta) \ln N$ but bounded. The input overlooked by that line of work is the 1999 theorem of Garcia and Tal. No claim is made on cycles, on the Effective Occupation Conjecture, or on the Collatz conjecture itself.

1 Introduction

Let $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n , acting on the positive integers, and for odd y let

$$U(y) = \frac{3y + 1}{2^{v_2(3y+1)}}$$

be the accelerated (Syracuse) map on odd integers. For an odd seed y_0 write $y_{n+1} = U(y_n)$, $a_n = v_2(3y_n + 1)$, $A_n = a_0 + \dots + a_{n-1}$, and

$$\lambda = \log_2 3, \quad g_n = n\lambda - A_n.$$

In the notation of De Jesús [5] and Rozier [4] the drift is $R_n = A_n - n\lambda = -g_n$; a divergent orbit must have $g_n \rightarrow +\infty$, and the question of how slowly g_n may grow measures how marginally

an orbit can diverge. Rozier's framework, and the occupation program built on it in [5, 6], produce unconditional results of the following shape: no divergent orbit satisfies $g_n \leq B \log_2 n$ eventually, for B below some threshold. The threshold reached in [5] is $5/6$, improved to $8/9$ by a mod 3 refinement, and Appendices D and E of [5] prove that $8/9$ is terminal for the congruence-packing method used there; Open Problem 12.8 of [5] states that any further progress requires a graded harmonic-thinness bound $\sum_{k < N} 1/y_k \leq (\frac{1}{3} - \eta) \ln N$ along divergent orbits.

The purpose of this note is to point out that a stronger statement than the one requested has been available since 1999. Garcia and Tal [1] proved that every infinite orbit of T has Banach density zero, by a pigeonhole argument over parity classes. Their theorem, made quantitative, gives a power-saving count of orbit points in every window, uniformly in the window position (Theorem 2.3 below, for which we give a short self-contained proof with an explicit exponent). From it: the reciprocal sum of a divergent orbit converges (Proposition 3.1), which is the harmonic-thinness requirement of [5] in the strongest possible form; the floor exclusion then holds for every $B \leq 1$ (Theorem 4.1); and a windowed occupation bound pushes it to every $B < 1/\beta^* = 1.03585\dots$ (Theorem 4.2, Corollary 4.3), past the harmonic barrier $B = 1$ that no packing argument over distinct states can cross.

Everything here is elementary. The only nonclassical ingredient is the observation of [1] that a windowed pigeonhole over parity classes applies to a single injective orbit; the counting that feeds it goes back to Terras [2] and Everett [3]. The exponent $\beta^* = r_H \log_2 3$, where r_H is the critical ones-ratio of Rozier [4], arises from the same entropy equation as his constant, which is reassuring rather than surprising: both express the tradeoff between the number of parity prefixes of a given tilt and the contraction they force.

Scope. Nothing below constrains hypothetical nontrivial cycles, proves any tier of the Effective Occupation Conjecture of [5], or approaches the Collatz conjecture. The contribution is the identification of [1] as the missing input to the occupation program, the explicit exponent, and the threshold $1/\beta^*$.

2 An explicit windowed sparsity bound

Throughout, $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ is the binary entropy. For $n \geq 1$ and $N \geq 0$ let $m_N(n)$ be the number of odd values among $n, T(n), \dots, T^{N-1}(n)$. Two classical facts [2, 3]:

Lemma 2.1. *The parity prefix $(T^j(n) \bmod 2)_{j < N}$ depends only on $n \bmod 2^N$, and $n \bmod 2^N \mapsto (\text{prefix})$ is a bijection of $\mathbb{Z}/2^N$ onto $\{0, 1\}^N$. Consequently, for every $r \geq 0$,*

$$T^N(n + r 2^N) = T^N(n) + r 3^{m_N(n)}.$$

Proof. Both statements follow by induction on N : an even step halves both the value and the offset, an odd step maps $n + r 2^j$ to $T(n) + 3r 2^{j-1}$, and in either case the step parity of $n + r 2^N$ agrees with that of n . $\square$

Lemma 2.2. *For all $n, N \geq 1$, $T^N(n) < 3^{m_N(n)}(2^{-N}n + 1)$.*

Proof. Writing $\varepsilon_j \in \{0, 1\}$ for the parity of $T^j(n)$ and unrolling, $2^N T^N(n) = 3^m n + E$ with $m = m_N(n)$ and $E = \sum_{j < N} \varepsilon_j 3^{m - m_{j+1}(n)} 2^j$. Each exponent $m - m_{j+1}(n)$ is at most $m - 1$ when $\varepsilon_j = 1$, so $E \leq 3^{m-1} \sum_{j < N} 2^j < 3^m 2^N$, and the claim follows. $\square$

Call a set $A \subset \mathbb{N}$ *collision-free* if distinct $u_1, u_2 \in A$ never satisfy $T^j(u_1) = T^j(u_2)$ for any $j \geq 0$. The value set of an infinite aperiodic orbit is collision-free: a collision between two of its points forces the orbit to become eventually periodic, hence bounded. This observation, and the window argument that follows, are due to Garcia and Tal [1]; we substitute explicit Terras–Everett counting for the theorem of Heppner used there, which is what makes the exponent effective.

Theorem 2.3. *Let A be collision-free and let $\gamma \in (\frac{1}{2}, 1/\lambda)$. Then for all integers $a \geq 1$, $X \geq 2$,*

$$\#(A \cap [a, a + X)) \leq 6(\lfloor \log_2 X \rfloor + 1) \left(X^{H(\gamma)} + X^{\gamma\lambda} \right).$$

In particular, with

$$\beta^* = \gamma^*\lambda = H(\gamma^*) = 0.9653844\dots, \quad \gamma^* = 0.6090897\dots$$

the unique root of $H(\gamma) = \gamma\lambda$ on $(\frac{1}{2}, 1/\lambda)$, one has, for every $\beta > \beta^$, a constant C_β with $\#(A \cap [a, a + X)) \leq C_\beta X^\beta \log(2X)$ uniformly in a .*

Proof. Put $N = \lfloor \log_2 X \rfloor$, so $2^N \leq X < 2^{N+1}$, and let $W = A \cap [a, a + X)$. Choose $z \geq 0$ with $z2^N < a \leq (z+1)2^N$; every $y \in [a, a + X)$ has $y - z2^N$ or $y - (z+1)2^N$ in $[1, X]$, so for one choice $z' \in \{z, z+1\}$ the set $B = \{s \in [1, X] : s + z'2^N \in W\}$ has $\#B \geq \#W/2$. Pigeonhole B on the value $m_N(s) \in \{0, \dots, N\}$: some $B_1 \subseteq B$ has a common value m and $\#B_1 \geq \#B/(N+1)$.

If $m \geq \gamma N$: by Lemma 2.1 the elements of B_1 occupy residue classes mod 2^N whose parity prefix has at least γN ones; there are at most $\sum_{i \geq \gamma N} \binom{N}{i} \leq 2^{NH(\gamma)}$ such classes (entropy bound, valid for $\gamma \geq \frac{1}{2}$), and each class meets $[1, X]$ in at most $X/2^N + 1 \leq 3$ points. Hence $\#B_1 \leq 3 \cdot 2^{NH(\gamma)} \leq 3X^{H(\gamma)}$.

If $m < \gamma N$: by Lemma 2.2, every $s \in B_1$ has $T^N(s) < 3^{\gamma N}(2^{-N}X + 1) \leq 3 \cdot 2^{\gamma\lambda N} \leq 3X^{\gamma\lambda}$, so T^N maps B_1 into a set of fewer than $3X^{\gamma\lambda}$ positive integers. If $\#B_1 > 3X^{\gamma\lambda}$, two distinct $s_1, s_2 \in B_1$ share $T^N(s_1) = T^N(s_2)$; since they share m as well, Lemma 2.1 gives $T^N(s_1 + z'2^N) = T^N(s_1) + z'3^m = T^N(s_2 + z'2^N)$, a collision between two distinct elements of A . Hence $\#B_1 \leq 3X^{\gamma\lambda}$.

Combining, $\#W \leq 2(N+1) \cdot 3(X^{H(\gamma)} + X^{\gamma\lambda})$. On $(\frac{1}{2}, 1/\lambda)$ the function H is strictly decreasing and $\gamma\lambda$ strictly increasing, so $\max(H(\gamma), \gamma\lambda)$ is minimized at the crossing γ^* , computed numerically to the stated precision. The equation $H(\gamma) = \gamma\lambda$ is, after dividing by γ , the equation defining Rozier’s critical ones-ratio r_H [4], whence $\gamma^* = r_H$. $\square$

Remark 2.4. Garcia and Tal [1] state Banach density zero; the power saving is implicit in their proof once Heppner’s exponents are made explicit, but we have not found the explicit exponent, or the identification $\beta^* = r_H\lambda$, in the literature. No claim is made beyond “no exact match identified.”

3 Reciprocal summability

Proposition 3.1. *Let $\mathcal{O}$ be the value set of a divergent orbit of T . Then $\sum_{x \in \mathcal{O}} 1/x < \infty$. Equivalently, since a divergent orbit visits pairwise distinct values, $\sum_{k \geq 0} 1/T^k(N_0) < \infty$ and, for the odd iterates, $\sum_{n \geq 0} 1/y_n < \infty$.*

Proof. A divergent orbit is aperiodic, so $\mathcal{O}$ is collision-free. Fix $\beta \in (\beta^*, 1)$ and sum Theorem 2.3 over dyadic blocks $[2^i, 2^{i+1})$: $\sum_{x \in \mathcal{O}} 1/x \leq \sum_{i \geq 0} C_\beta 2^{i\beta} (i+2) 2^{-i} < \infty$. $\square$

This settles the requirement of [5, Open Problem 12.8], which asks for $\sum_{k < N} 1/y_k \leq (\frac{1}{3} - \eta) \ln N$ for some $\eta > 0$ along divergent orbits: the sum is in fact bounded. In the carry-budget notation of [5], $E_N = \sum_{k < N} \log_2(1 + \frac{1}{3y_k})$ is bounded along every divergent orbit, rather than of size $\frac{1}{6} \log_2 N$.

4 Floor exclusion and occupation

The product form of the orbit identity (Eliahou; Rozier [4]; see also [5]) reads, for the odd iterates of any orbit,

$$y_n = y_0 2^{g_n} Q_n, \quad Q_n = \prod_{j < n} \left(1 + \frac{1}{3y_j}\right) \text{ nondecreasing.} \quad (1)$$

For a divergent orbit, Proposition 3.1 gives $Q_n \nearrow Q_\infty < \infty$, so

$$y_0 2^{g_n} \leq y_n \leq y_0 Q_\infty 2^{g_n} \quad (n \geq 0). \quad (2)$$

Theorem 4.1. *No divergent orbit satisfies $g_n \leq B \log_2 n + K$ for all large n with constants $B \leq 1$, $K \geq 0$.*

Proof. By (2) the hypothesis gives $y_n \leq y_0 Q_\infty 2^{K n^B}$ for large n , so $\sum_n 1/y_n \geq c \sum_n n^{-B} = \infty$ for $B \leq 1$, contradicting Proposition 3.1. $\square$

Theorem 4.2. *Let (y_n) be the odd iterates of a divergent orbit and fix $\beta \in (\beta^*, 1)$. There are constants c_1, c_2 depending on β and on the orbit (through y_0 and Q_∞) such that*

$$\#\{n \geq 0 : g_n \leq G\} \leq c_1 2^{\beta G} (G + c_2) \quad \text{for all } G \geq 1.$$

Proof. If $g_n \leq G$ then $y_n \leq y_0 Q_\infty 2^G$ by (2). The y_n are pairwise distinct elements of the collision-free set $\mathcal{O}$, so by Theorem 2.3 with $a = 1$, $X = y_0 Q_\infty 2^G$, the number of such n is at most $C_\beta (y_0 Q_\infty)^{\beta} 2^{\beta G} \log(2 y_0 Q_\infty 2^G)$. $\square$

Corollary 4.3. *No divergent orbit satisfies $g_n \leq B \log_2 n + K$ for all large n with $B < 1/\beta^* = 1.0358567\dots$.*

Proof. Suppose it did, with $B > 0$ (for $B \leq 0$ apply Theorem 4.1). For large G , every $n \leq 2^{(G-K)/B}$ beyond the threshold has $g_n \leq G$, so $\#\{n : g_n \leq G\} \geq \frac{1}{2} 2^{G/B} 2^{-K/B}$. Against Theorem 4.2 this forces $1/B \leq \beta$ for every $\beta > \beta^*$, i.e. $B \geq 1/\beta^*$. $\square$

In the drift notation of [5] ($R_k = -g_k$): no injective accelerated orbit satisfies $R_k \geq -B \log_2 k$ eventually, for any $B < 1.0358\dots$, extending the thresholds $5/6$ and $8/9$ of [5, Theorem C, Appendix D] past the harmonic barrier $B = 1$, which no packing over distinct states can cross ([5, Appendix E] proves $8/9$ terminal for fixed-modulus congruence packing; the present input is of a different, windowed, kind).

5 Limitations

The exponent β^* is a genuine barrier for the argument of Theorem 2.3: it balances the entropy of heavy parity prefixes against the contraction of light ones, and both sides are sharp for the hypothesis class. Improving $1/\beta^*$ therefore requires either deeper iteration than $N \approx \log_2 X$ inside the window argument, which breaks Lemma 2.1's modulus bookkeeping, or an input beyond parity counting. Nothing here constrains nontrivial cycles: a periodic orbit is not aperiodic, so it lies outside every hypothesis above, and the theorems concern divergence candidates only, exactly as in [5, Remark 7.4]. Nothing here bears on the sub-critical occupation of convergent orbits, which is the open core of the Effective Occupation Conjecture: in the language of [6], the anti-concentration problem for the residue classes of near-critical words is untouched.

Acknowledgment

This note exists because the occupation program of De Jesús [5, 6] states its open problems precisely enough to be answered by a literature search.

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